# Aryan Patel

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## RESEARCH INTERESTS

Deep Learning Systems & Architectures · Foundation Models · Scientific Computing · Data Analysis

## EDUCATION

**Jabalpur Engineering College** — Jabalpur, MP
*Bachelor of Technology – Artificial Intelligence and Data Science* — *Aug 2025 – Jun 2029*
*Standing:* Second Year (Sophomore)

**Stemfield International School** — Jabalpur, MP
*Higher Secondary Education – PCM + Computer Science (CS)* — *Apr 2011 – Mar 2025*

## RESEARCH PROJECTS

**NeuralFabric: Machine Learning Framework from Scratch** | *Python, NumPy*

- Built a modular machine learning framework from first principles using Python and NumPy, implementing tensor operations, reverse-mode automatic differentiation, and foundational regression models.
- Developed numerical-gradient tests with Pytest to validate autograd correctness, enforced static typing with MyPy, and structured the project for open-source distribution with modern Python packaging and GitHub Actions.

**AdityaL1-MAG: Solar Magnetic Field Analysis** | *Python, Signal Processing*

- Developed a data-analysis pipeline for ISRO's Aditya-L1 MAG observations, including data ingestion, filtering, and visualization of spacecraft magnetic-field measurements.
- Performed magnetic-field characterization and exploratory analysis to identify patterns and variations in the observed environment.

**Timeless: Decoder-Only Transformer Language Model** | *PyTorch, Transformers*

- Built a decoder-only Transformer language model from scratch in PyTorch, implementing tokenization, causal self-attention, multi-head attention, residual connections, and normalization.
- Developed an end-to-end LLM workflow covering pretraining on unlabeled text, model checkpointing, text-classification fine-tuning, and instruction tuning for downstream tasks.

**Diagnostic Neural Network for Oncology** | *TensorFlow, Scikit-learn*

- Built a TensorFlow neural network for breast cancer classification using standardized clinical features and a structured preprocessing pipeline with Scikit-learn.
- Evaluated model performance using accuracy, confusion matrix, and classification metrics, achieving ∼94% test accuracy, and saved the trained model for reuse.

## CERTIFICATIONS

**Machine Learning Specialization** | *DeepLearning.AI, Stanford University* — Jul 2026

- Completed coursework in supervised and unsupervised learning, neural networks, decision trees, model evaluation and optimization, regularization, clustering, anomaly detection, recommender systems, and reinforcement learning.

**Mathematics for Machine Learning (Gold + Elite)** | *NPTEL, IIT Kharagpur* — Nov 2026

- Achieved 91%, earning the Gold + Elite Badge in Mathematics for Machine Learning, offered by the Indian Institute of Technology, Kharagpur through NPTEL.

**Python for Data Science (Elite + Silver, Top 2% Topper)** | *NPTEL, IIT Madras* — Oct 2026

- Earned the Elite + Silver certification with an 85% score, ranking among the top 2% of candidates nationwide in NPTEL's Python for Data Science course offered by IIT Madras.

## AWARDS & HONOURS

**Finalist, RAGNAROK Hackathon** | *Google Developer Groups (GDG)* — Jan 2026

- Selected as a finalist in a competitive hackathon for technical problem-solving and rapid project development.

## TECHNICAL SKILLS

**Programming Languages:** Python, C, C++, Java, Go, R, JavaScript, TypeScript
**Machine Learning:** PyTorch, TensorFlow, XGBoost, Scikit-learn
**Scientific Computing:** NumPy, Pandas, Matplotlib, Scientific Data Analysis
**System & Tools:** Linux, Bash, Git, GitHub